THE COMPLETE PROJECT GUIDE · PLAIN-LANGUAGE EDITION

AUS Backyard Intelligence

A smart map of Australia that finds swimming pools and solar panels from the sky — and turns them into useful, lawful business leads. This is the full story: every feature, every design decision, every experiment, explained simply.

Build r22 · July 2026 · No jargon left unexplained — there is a dictionary at the end.

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1. What is this project?

Imagine looking at Australia from above and being able to ask: *"Which houses have a swimming pool? Which roofs have solar panels? And which backyards would be perfect for a new pool, deck, or fence?"* AUS Backyard Intelligence answers those questions. It is a web application that combines satellite photos, free public map data, and two artificial-intelligence "eyes" that scan images and recognise pools and solar panels — then presents everything on an interactive map that anyone can use in a browser.

Everything runs on one ordinary computer. There are no cloud AI fees, no paid data subscriptions, and no accounts to create. Every data source is free and public, and every accuracy claim in this document comes from a repeatable, automated test.

> **Running it takes four steps.** ① Create a Python environment and run `pip install -e .` in the project folder. ② Start the server: `python -m aus_backyard_intelligence --port 8048`. ③ Open `http://127.0.0.1:8048` in a browser (the leads dashboard is at `/dashboard`, the 3D view at `/3d`). ④ Type a suburb, press Scan, and export the results as a spreadsheet.

2. Who is it for?

The main users are **tradespeople and businesses that work on backyards**:

- **Pool builders** — find streets full of pool-friendly backyards that don't have pools yet.
- **Pool-safety certifiers** — find pools that are visible from the sky but may be missing from the state safety register (QLD and NSW keep public ones). Each of those is a potential inspection job.
- **Fencing, deck and landscaping companies** — every property gets suitability scores for fencing, decks and landscaping, not just pools.
- **Solar installers** — see which roofs already have panels and which sunny roofs don't.
- **Anyone curious** — it is also simply a fascinating map to explore.

> **A real example.** You type "Broadbeach Waters" into the search box and press Scan. A few minutes later the map shows every detected pool in the area, a ranked list of promising properties with street addresses, links to the official land-titles registry and the state pool register, and a button to export everything as a spreadsheet for your sales team.

3. A tour of the app, screen by screen

3.1 The map viewer (the home page)

Open `http://127.0.0.1:8048` and you get a full-screen map. Around it:

- **Top-left — search box.** Type any suburb or address; suggestions appear as you type (powered by OpenStreetMap's free address service). Press *Locate* to jump to your own position. The app remembers where you were and reopens there next time.
- **Left panel — the intelligence panel.** A suburb scan box with a "passes" setting (how many sweeps the deep scan should make), live counters (**Pools · Solar · Sites · Leads**), a running status line ("Realtime YOLO: 18 pools"), an *Open leads dashboard* button and an *Export detections (CSV)* button.
- **Right panel — layers and controls.**
  - **Base map**: Google Satellite, Google road map, or *AU Government Aerial (all Australia)* — one seamless layer that automatically picks the right state government's official photography under every tile and falls back to Google where none exists.
  - **View perspective**: standard 2D or a tilted pseudo-3D view.
  - **Spatial overlays** (tick boxes): AI Canopy (vegetation), Solar Panels (AI detected), Roof Solar Suitability, Drainage Contours.
  - **Local feature detection** (tick boxes): Pools / Aquatic venues, Structure Footprints, Vegetation Canopies, 3D Structure Meshes.
  - **Select analysis area** and **Run area scan** buttons with an "auto scan selected areas" option.
- **Bottom-left — the build badge.** A small label like `build r22 · ensemble · recall 93% · precision 92%`. It exists so you always know which version your browser tab is actually running — a stale tab is the oldest trick in the book for "the fix didn't work".

3.2 What you see on the map

- **Pools** appear as red outlines (or dots when zoomed out). Click one and a popup shows what found it ("Mapped pool" from OpenStreetMap, or "Pool-trained YOLO detection"), the confidence percentage, an *Open in Google Maps* link, and the correct state's pool-register link.
- **Solar arrays** appear as cyan outlines with size (m²) and confidence in the tooltip.
- **Buildings** appear as pink outlines with their type and floor area.
- **Trees/canopy, drainage contours and 3D meshes** appear when their overlays are ticked.
- While you simply pan around, a **background heartbeat** keeps scanning whatever you're looking at (at zoom 15 or closer), so pools appear "by themselves" within seconds.

3.3 Analysing one property or area

Press *Select analysis area*, draw a shape around a backyard (or a whole block), and the analysis panel opens with:

- **Suitability ratings** — scores out of 100 for Swimming Pool, Pool (flat), Fencing, Deck / Patio and Landscaping, each with a coloured bar and a verdict word.
- **Property metrics** — total area, backyard area, backyard depth, street frontage, side access width, slope, elevation and vegetation.
- **A written AI assessment** — a plain-English summary: what the site is like, the "bottom line", and how confident the system is.
- **Quick external views** — one-click Street View, 3D view, Google Earth and Google Maps.
- **Export to QGIS** — downloads the analysis as a professional GIS file (GeoPackage).
- If the area already contains a pool, the app says so and switches new-pool scoring off — an existing pool means renovation and compliance leads, not construction leads.
- **Manual corrections** — you can mark a pool the AI missed by hand, or clear wrong markings, directly on the map.

3.4 The leads dashboard

`/dashboard` is a second page for the business side: a ranked table of properties (address, suburb, score, explanation, backyard metrics), the background-scan status, metric tiles (lead count, top score, leads with published contact), and per-lead cards with the evidence (footprint size, pools nearby, green features, powerlines) plus the full set of lawful lookup links — Maps, Street View, realestate.com.au, Domain, land titles, **pool register**, council planning and ABN lookup for businesses.

3.5 The 3D page

`/3d` reconstructs an area in three dimensions — terrain, extruded buildings and up to ten thousand trees rendered efficiently — useful for visualising slope and shading around a property.

4. How pool detection works, step by step

This is the heart of the project. Here is the full journey from "satellite photo" to "pool on your map", with nothing skipped:

1. **Choose the zoom.** The detector always tries to work at the sharpest zoom level (20 — roughly 15 cm per pixel). This matters enormously: at zoom 18 the AI found only 10 of 55 pools in a test cell; at zoom 20 it found 48. An early version of the app accidentally scanned at whatever zoom your *screen* was using — fixing that one line more than doubled live detections.
2. **Cut the area into chunks.** The viewport is divided into overlapping tile groups sized for the AI (about 1024 pixels square). Big areas get a "budget" so a scan can't run forever; deep scans get a much larger budget than live browsing.
3. **Stitch the tiles.** Individual 256-pixel map tiles are downloaded and stitched into one image per chunk. Failed tiles are recorded — the app never pretends it saw an area it didn't.
4. **Inspector A looks.** The base AI model scans the chunk and outputs boxes: "pool here (61%), pool here (88%), solar panel here (74%)...". Anything below 15% confidence is discarded — that floor was chosen by measuring the full accuracy curve, not by guessing (see section 9).
5. **Inspector B looks.** The second, Australian-trained model scans the same chunk for pools. Any pool it finds that Inspector A didn't is added to the pile (details in section 5).
6. **The solar veto.** Solar panels are the number-one pool impostor from above (dark rectangles, bluish shimmer). Any "pool" whose centre sits inside a detected solar array is discarded.
7. **Colour and context checks.** Every surviving box gets its pixels examined: how much of it looks like water? Is it suspiciously roof-coloured? In live (fast) mode these checks are strict, because a single glance has no second chance. In deep-scan mode they are more forgiving, because the multi-pass voting (section 6) catches mistakes another way — that one change recovered dozens of tree-shaded and dark-bottomed pools that used to be rejected.
8. **Size gates.** Backyard pools are typically 8–400 m². Anything outside that is rejected — *unless* it passes the "large clear water" test (huge, almost entirely water, strongly blue), which is how Olympic and public pools get through. Solar arrays have their own range (2–2,500 m²).
9. **Pixels become coordinates.** Each box is converted from pixel positions to real latitude and longitude, its real-world area is estimated, and (where possible) the actual water outline is traced rather than just a rectangle.
10. **Merging with the public map.** Pools already drawn in OpenStreetMap are trusted and shown as "Mapped pool". If the AI detects a pool within 20 m of a mapped one, the two are merged so the map shows each pool exactly once — but the merge *records* that the AI independently confirmed it (this little bookkeeping note turned out to be critical for honest accuracy measurement — section 9).
11. **Delivery.** The finished list — mapped pools, new AI pools, solar arrays — flows to the map, the counters, the CSV export and the lead scorer. The client keeps a local cache so revisiting an area is

instant, and the server keeps its own short-term cache to avoid rescanning the same view.

> **Diagnostics all the way down.** Every scan records what happened: how many tiles loaded, how many candidates each stage rejected and *why* (with per-reason counters), and how long it took. When something looks wrong, the answer is in the diagnostics, not in guesswork.

5. The two-inspector ensemble

The pool detector is not one AI — it is two, working as a team.

Inspector A — the generalist

A published, pool-trained YOLO model (v8-large, "P2" variant tuned for small objects, trained at high resolution on satellite imagery, with two classes: *pool* and *solar panel*). It was chosen by a bake-off: three candidate models were benchmarked through the identical production pipeline, and this one found roughly 60% more pools than the previous default.

Inspector B — the local

A copy of Inspector A that we re-trained on **929 Australian photo chips** built from public map data: Gold Coast canal suburbs, Brisbane's north, and Perth's western suburbs — three different looks (different roofs, soil colours, light). Two details of that training matter:

- **Every pool in every photo was labelled** — not just the one in the middle. An unlabelled pool in a training photo silently teaches the AI to *ignore* pools. Dense suburbs cluster pools, so this fix mattered.
- **The solar skill was protected.** The training photos were full of solar panels that public maps don't record. Training on them unlabelled would have destroyed the model's solar detection. So the current model first labelled the solar panels itself (6,221 of them), and those labels were kept during training — a standard trick called self-distillation.

Why two are better than one

Tested alone, Inspector B was actually *slightly worse* at finding pools (we rejected it as a replacement and wrote the failure down). But its mistakes were **different**: it recovered 22 of the 35 pools Inspector A missed, while adding zero new false alarms. So the app keeps everything either inspector finds, with duplicates merged. Result: pool detection jumped from 87% to 94% overnight. The training areas were deliberately chosen far from the accuracy-test area (Sydney's east), so the exam could never be "leaked" into the study material.

6. Deep area scans and how voting works

Live browsing gives one quick look. The *Run area scan* button is the thorough version:

1. The chosen area (drawn by hand, or a suburb by name) is scanned **N times** — a full baseline sweep plus sweeps shifted by roughly half a tile, plus quadrant close-ups for large areas. Pools that sit on a tile seam in one alignment sit in the middle of a tile in another.
2. Detections from all passes are **clustered**: boxes within 7 m of each other are treated as the same pool, and the app tallies votes — how many passes saw it, out of how many passes that actually covered that spot.
3. Each pool gets a **verdict**: *mapped* (from OpenStreetMap — always kept), *confirmed* (seen repeatedly or with very high confidence), *probable*, or *rejected* (seen once, missed by several other passes that looked right at it).
4. Scans run **in the background** on the server with a job queue, progress reporting, and a cap on simultaneous jobs — you can keep browsing while a 12-pass suburb scan grinds away.
5. Confirmed pools carry their voting record (`passes_seen` , `coverage` , `agreement`) so you can always ask "how sure are we?" about any single pool on the map.

7. Leads: from detection to business

Detections are interesting; leads pay the bills. Here is the full pipeline:

1. **Candidate selection.** After a scan, every *residential* building (house-like type, or a generic building with a house-sized footprint of 45–900 m²; sheds and shops excluded) with **no pool on its own land** (nothing within 22 m) becomes a candidate.
2. **Scoring.** Candidates earn points for a house-sized footprint, for **neighbourhood pool density** (pools within 180 m but not on the lot — the single best "people here buy pools" signal), and for moderate greenery (a real yard); they lose points for overhead powerlines (a crane problem and a safety rule problem).
3. **Address resolution.** The top leads are reverse-geocoded to real street addresses (rate-limited and cached, respecting the free service's rules), and duplicates collapse to the best-scoring lead.
4. **Legal parcel.** In QLD and NSW, the lead is matched to its official cadastral parcel — the lot/plan number a certifier or conveyancer actually uses.
5. **Jurisdiction stamping.** Every lead is stamped with its state and that state's pool-compliance rules (see section 10) — the dashboard link always points at the *right* register.
6. **Lawful contact only.** If the property itself published contact details on the public map (mostly businesses do this), they are shown. Otherwise the lead links to the state land-titles registry, council planning search, realestate.com.au, Domain, and ABN lookup. The app **never** invents, guesses or scrapes personal details (more in section 11).

7. **Hand-off.** Leads land in the dashboard and the CSV export — typed, scored, addressed, stamped and linked, ready for a CRM.

8. The self-improving data engine

Switch on **Dev mode** and the app becomes its own data collector:

- **One-click truth.** Every pool popup gains ✓ Correct / ✗ Not-a-pool buttons. Answers are stored permanently with position and context — corrections become tomorrow's training data, and ✗ answers immediately store a "no pool here" ground-truth point.
- **Labels while you browse.** Every new neighbourhood you settle on (roughly 900 m cells) is automatically harvested once: mapped pools become positive labels, pool-free houses become negative labels, mapped solar becomes solar labels. Zero clicks.
- **Training candidates from scans.** When a deep scan finishes, its scan-*confirmed* pools (seen 2+ times, decent confidence) and confident solar arrays are appended to a training-candidate file — with duplicates filtered. This file is exactly what fed Inspector B.
- **The permanent exam.** The evaluation harness keeps a labelled ground-truth store (**581 points** currently: 258 known pools, 323 known pool-free spots in Sydney's east) and can re-score the entire production pipeline on demand, appending every result to a history file. A strict rule keeps it honest: **machine-generated guesses never enter the exam's answer key** — ground truth comes only from the public map and human feedback.
- **Status at a glance.** The dev panel shows the pipeline's running totals — labels by type, training candidates by type, feedback count — plus one-click export of the training file.
- **Production switch.** One setting (`ABI_DEV_API=0`) turns all of this off for a public deployment.

9. The accuracy story — every experiment, in order

Accuracy work here follows one discipline: **measure, change one thing, measure again, write it down** — including the failures. The full log lives in `docs/accuracy_log.md`; this is the complete story in plain words.

9.1 First, the measuring stick itself was bent — twice

- **The southern-hemisphere bug.** The exam grid put every label one cell north of where it really was (a classic negative-number rounding mistake that only bites below the equator). The system was being tested on the wrong squares of the map. Found, fixed, regression-tested.
- **The Overpass dependency.** Worse: the score changed depending on whether a third-party map service was busy. The de-duplication step was silently hiding the AI's detections of already-mapped pools — precisely the pools the exam asks about. When that service was healthy the score read 35%; when it was rate-limited the same code scored 89%. The fix: the merge now leaves a note ("the AI independently found this one") that the exam counts. The score became stable regardless of third-party weather, and three tests make sure it stays that way.

9.2 Then, tuning — measured, not guessed

With a trustworthy ruler, the confidence floor (how sure the AI must be before we accept a box) was swept across the whole range:

Floor	Pools found	False-alarm rate	Overall balance (F1)
0.20	86.4%	12.7%	0.854
0.15 (chosen)	**87.2%**	**13.0%**	**0.857 — the peak**
0.12	88.0%	14.2%	0.855
0.10	88.0%	14.2%	0.855

Below 0.12 nothing more appears — the single model's ceiling was ~88%. Every missed pool was then examined individually: for 10% of pools, the AI had *no idea at all* even at the most generous setting — they are under trees or inside enclosures. No amount of tuning fixes blindness; only a better model could.

9.3 The fine-tune that "failed" — and won anyway

Model (production test)	Pools found	False-alarm rate
Inspector A alone	87.2%	13.0%
Inspector B alone	85.7% — **rejected** as a replacement	11.5%
Both together (build r21)	**93.8%**	13.0%

The pre-registered rule said a new model ships only if it beats the old one on finding pools. B alone didn't — that rejection is in the log. But B's misses were different from A's, and the union cleared 90% for the first time. Scan time grew only ~1.4× (downloading tiles, not AI, dominates the clock).

9.4 The precision pass (build r22 — current)

With recall in the target band, attention turned to false alarms. Every one of the 42 false positives was traced to its source: **28 came from an older colour-based detector** — almost all of them the *ocean* near clifftop houses, which genuinely is clear blue water — and only 14 from the AI ensemble. That colour detector predates the ensemble and existed to catch bright obvious pools the old single model missed; the ensemble now catches those itself, leaving the colour detector with 10 unique finds against 28 false alarms. It now switches itself off automatically whenever the second model is present.

Build	Pools found (recall)	False-alarm rate	Right when it says "pool" (precision)	Balance (F1)
r21 ensemble	93.8%	13.0%	85.2%	0.893
r22 (current)	**92.6%**	**6.5% — halved**	**91.9%**	**0.923 — best ever**

> **One more honesty note.** The "false alarms" are counted against pool-free spots derived from the public map — and the public map is incomplete. Spot-checks show some counted "mistakes" are real pools nobody ever drew on OpenStreetMap. True precision is therefore *better* than 92%; we quote the pessimistic number.

10. Australia-wide coverage and the state rule book

10.1 One seamless national imagery layer

Five states publish free, high-quality aerial photography. The app's "AU Government Aerial" basemap routes every single map tile to the right source automatically:

Region	Imagery source	Detail level
Queensland	QLD Government Latest State Program	zoom 20
NSW + ACT	NSW Spatial Services (SIX)	zoom 20
Victoria	Vicmap Aerial	zoom 20
South Australia	Location SA Public Mosaic	zoom 21
Tasmania	theLIST Orthophoto	zoom 19
WA, NT, remote areas, ocean	Google satellite (automatic fallback)	—

Border handling is deliberate: the Victoria/NSW border follows the Murray River, Queensland's south-east corner follows the Tweed — the router approximates both so border towns get the right state's photos. Every service above was probed and verified tile-by-tile before shipping; WA, NT and the ACT simply don't publish free statewide tiles (WA's is subscription), so Google covers them.

10.2 The pool-compliance matrix

Pool-safety law is state law — there is no single national rule. The app carries the whole matrix, and every property is matched to its state (with the official cadastre overriding the geometric guess near borders):

State	How pool compliance works there	What the app links to
Queensland	Every regulated pool must appear in a **public state register**; a safety certificate is needed to sell or lease.	QBCC pool safety register (searchable)
New South Wales	Registration mandatory on a **public state register**; compliance certificate needed to sell or lease.	NSW Swimming Pool Register (searchable)
Victoria	Pools/spas must be registered with the **local council**; barrier certified every four years.	State building regulator's pool pages
Western Australia	No public register; councils must inspect barriers at least every four years.	Building and Energy pool-safety rules
South Australia	Barrier rules under planning law; councils inspect new pools.	PlanSA
Tasmania	Barriers under the Building Act 2016, council-enforced.	CBOS

ACT	Barrier and registration requirements via the territory regulator.	Access Canberra
Northern Territory	Compliance certificates under the Swimming Pool Safety Act, required on sale/lease in regulated areas.	NT pool-safety pages

For a certifier, the QLD and NSW rows are gold: a pool visible from the sky but absent from a public register is a concrete, checkable lead. The same state table exists twice — once in the server, once in the browser — and an automated test fails the build if the two copies ever drift apart.

11. Privacy, law and security — in detail

11.1 Privacy by design

- **No invented people.** An early demo dataset shipped with made-up owner names, emails and mobile numbers ("Robert Thomas, 0410 931 258"). This was found and completely removed — a randomly generated mobile number is very likely *someone's real number*. The code that generated fake owners now returns empty contact fields by design, and an automated test fails the build if personal data ever reappears in the dataset.
- **No skip-tracing.** The app never attempts to discover who owns a property. Owner follow-up goes through the front door: the state land-titles registry (linked on every lead, all 8 jurisdictions), which is the lawful, auditable route.
- **Published contacts only.** The only contact details shown are ones the occupier published themselves on the public map (typically businesses), and website links are only rendered if they are ordinary web addresses.
- **Direct-marketing rules respected.** The design follows the Australian privacy principle for direct marketing (APP 7): transparency about sources, no inferred personal data, lawful lookup flows.

11.2 Security hardening (from a dedicated review pass)

- **File-serving containment.** The web server mathematically verifies every requested file lives inside the public folder — the classic "../" path trick and a subtler sibling-folder trick are both dead ends, verified by live probes.
- **Script-injection (XSS) defence.** Anything user-editable that reaches the page — including community-edited map tags like a building's type or a business's website — is escaped or scheme-checked so it can never run as code. (OpenStreetMap is wonderful, but anyone can edit it; the app treats its text as untrusted.)
- **Browser hardening headers** on every response, JSON body-size limits, timeouts on every outbound call, and thread-locks around all shared files.
- **Local by default.** The server binds to your own machine (127.0.0.1) unless you explicitly choose otherwise, and the dev/evaluation APIs can be disabled with one switch.

- **Third-party model caution.** Downloaded AI weights are in a code-executing format (pickle); before first use they were statically scanned opcode-by-opcode and approved by a human.

12. The data sources (all free)

Source	What the app uses it for
Google map tiles	The default satellite view and the imagery the AI scans while you browse.
State government aerial imagery (×5)	The seamless AU Government Aerial basemap (section 10.1).
OpenStreetMap (via Overpass)	Mapped pools (trusted ground truth), buildings, trees, powerlines, published business contacts, and the bootstrap labels for the accuracy exam.
Nominatim (OpenStreetMap)	Place search, autocomplete, and turning coordinates into street addresses — rate-limited and cached to respect the free service.
Microsoft Building Footprints	~11 million Australian building outlines, downloaded on demand per region and cached — fills OpenStreetMap's gaps so lead generation works in sparsely-mapped suburbs (one test suburb went from 0 leads to 18 purely from this).
QLD DCDB & NSW SIX cadastre	Official lot/plan parcel lookup for leads.
Elevation service	The terrain grid behind slope, drainage contours and the 3D view.
Public safety feeds (extras)	Optional map extras: live ISS position, recent earthquakes, flights, wildfires, traffic cameras and incidents — situational-awareness garnish, not core.

13. Under the hood: the parts of the machine

For the technically curious — each module in one honest sentence:

Part	What it is
`app.py`	The web server. Plain Python (no heavy framework), threaded so a slow scan never blocks the map, serving the UI and ~30 API routes with the security measures of §11.
`yolo_detector.py`	The detection engine — tiles, stitching, zoom logic, both AI models, the ensemble merge, colour/water/size gates, solar handling, and rich diagnostics. The biggest file, because it is the product.
`external_api.py`	The data hub — talks to OpenStreetMap, Nominatim and the elevation service; merges and de-duplicates everything into one clean feature list; caches aggressively.
`scan_agent.py`	The deep-scan machinery — background jobs, multi-pass voting, lead scoring, address enrichment, lawful links.

`evaluation.py`	The exam room — label store, automatic label bootstrap, the benchmark runner, feedback storage, training-candidate harvesting.
`jurisdiction.py`	The state rule book — point-to-state resolution and the compliance matrix (mirrored in the browser, drift-guarded by a test).
`cadastre.py` / `ms_buildings.py`	Lot/plan lookup; Microsoft footprints on demand.
`map_export.py` / `eligibility.py`	Property scoring, the written assessments, QGIS exports; the private-residential filter.
`static/`	The browser side — the map viewer, dashboard and 3D page, written in plain JavaScript with Leaflet (the standard open-source map library). No build step, no framework.
`scripts/`	The training kitchen — dataset builder (2-class, every-pool-labelled chips) and the retraining pipeline that produced Inspector B.
`tests/` (117 tests)	Automated checks for every module, including regression tests for every bug this document mentions.
`models/`	The two AI checkpoints (~50 MB each) plus the previous-generation model kept as a rollback.
`eval_data/` · `docs/accuracy_log.md`	The exam's answer key, its full history, the feedback file, the training candidates — and the experiment log this document draws from.

14. Settings and switches

Everything important can be controlled without touching code (the `ABI_` prefix is current; older `QLDR_` names still work):

Setting	Default	What it controls
`ABI_DEV_API`	on	Set to 0 to disable all dev/eval/feedback endpoints in production.
`QLDR_POOL_YOLO_CONF`	0.15	The AI confidence floor (the measured F1 peak).
`QLDR_POOL_YOLO_ENSEMBLE`	on	Set to 0 to run only Inspector A.
`QLDR_POOL_YOLO_WEIGHTS` / `_B`	models/*.pt	Point either inspector at different weights files.
`QLDR_POOL_YOLO_IMGSZ`	auto	AI input resolution; auto reads the model's own training size.

`QLDR_ENABLE_STRICT_WATER_SUPPLEMENT`	auto	The legacy colour detector: auto = only when the ensemble is absent; 1/0 force on/off.
`QLDR_ENABLE_CV_POOL_FALLBACK`	off	An even older colour-only detector kept for experiments (measured ~0% precision — stays off).
`QLDR_EVAL_DIR`	eval_data/	Where the exam's labels, reports and feedback live.
`QLDR_MS_BUILDING_CACHE` / `QLDR_DISABLE_MS_BUILDINGS`	temp / on	Microsoft footprint tile cache location; kill-switch.

15. The service directory (API)

Everything the browser does goes through ordinary web endpoints, so other software can use the engine directly:

Endpoint group	What it offers
`/api/health`	Is the server alive and what version is it.
`/api/osm/features`, `/api/local/features`	The core: give it a bounding box, get back pools, solar, buildings, trees (with live AI detection).
`/api/analyze/area`	Full property analysis for a drawn polygon (scores, metrics, written assessment).
`/api/agent/scan-place`, `/api/agent/scans`, `/api/agent/leads`	Start deep scans by suburb name; poll progress; fetch ranked leads.
`/api/rankings`, `/api/viewer/property/…`	Ranked demo properties and per-property detail.
`/api/terrain/elevation-grid`	Elevation matrix for any small area.
`/api/geocode`	Place search.
`/api/export/qgis…`	GeoPackage downloads.
`/api/eval/…`, `/api/feedback/…`	The data engine (labels, benchmark runs, reports, feedback, training export) — gated by the production switch.
`/api/osint/…`	The situational-awareness extras (ISS, quakes, flights, fires, cameras, incidents).

16. Build history, limits, roadmap — and the dictionary

16.1 How the project grew (abridged build history)

Build	What changed
r8	Fixed an invisible-map bug on high-resolution screens (a graphics-card texture limit); added the build badge.
r11	The accuracy exam was born (label store + benchmark + feedback buttons); cadastre lookup; pool outlines.
r12	Three pool models benchmarked head-to-head; the best one adopted (+60% more pools found).
r13	Five state government aerial basemaps, each probed and verified tile-by-tile.
r14	Two big fixes: detection stopped inheriting the *screen's* zoom (live pools 112 → 242), and the combined all-Australia basemap shipped. The southern-hemisphere exam bug was found and fixed.
r15–r16	Solar panels became first-class detections; solar counter and CSV export.
r17	The automatic data pipeline (labels while browsing, training candidates from scans); confidence floor tuned by measurement.
r18	UI simplification: one combined basemap choice, split solar toggles, instant overlay rendering.
r19	**The national rebrand**: package renamed, jurisdiction rule book for all 8 states, fake owner data deleted, 40 accidentally-American demo records deleted (a suburb-name lookup had matched "Warner", New Hampshire), dev-API production switch, wheel packaging fixed.
r20	Whole-code cleanup (no emojis, no filler), linting, dead code removal — plus a security review: path containment, XSS escapes, link scheme checks, hardening headers.
r21	**The two-model ensemble** — pool recall 93.8%.
r22	**The precision pass** — false alarms halved, precision 91.9%, best F1 ever (0.923). Current build.

16.2 Known limits (the honest list)

- Pools fully hidden by trees or roofs cannot be seen in any overhead photo (~7% of pools).
- Free satellite imagery can be one to three years old; a brand-new pool may not appear yet.
- Solar-panel detection works well visually, but its exact accuracy cannot be scored yet — public maps rarely record rooftop solar, so there is no answer key; human spot-checks are the next step.
- WA, NT and the ACT lack free state aerial imagery (Google fills in).
- Roof condition scoring and construction-progress tracking (features of paid competitors with their own aircraft) need imagery this project does not have — out of scope for the free version.

- Accuracy numbers move about ±1% between identical runs (tile downloads are not perfectly repeatable); differences smaller than that are noise, and the log says so.

16.3 Roadmap (next sensible steps)

- **Human labels for the 21 remaining false alarms** — several are probably real unmapped pools; ten minutes of ✓/✗ clicking would sharpen both the map and the exam.
- **G-NAF address validation** (the national address file) for authoritative addresses.
- **National elevation (ELVIS)** for better slope and drainage everywhere.
- **A licensed-imagery adapter** — the detection engine is imagery-agnostic by design; plugging in paid high-resolution photography (e.g. 5–7 cm aerial) would directly attack the tree-hidden misses and freshness limits.
- **A GPU training run** — Inspector B was trained on an ordinary CPU overnight; proper GPU training on the growing training-candidate file is the next model-quality jump.
- **Richer scoring** — splitting the single score into Opportunity, Buildability, Compliance Risk, Data Confidence and Estimated Job Value.

16.4 Little dictionary

Term	Plain meaning
YOLO	"You Only Look Once" — an AI that finds objects in a photo in a single pass. Fast and accurate.
Recall	Of all the real pools out there, the share we actually found. Ours: 93%.
Precision	When we say "pool", the share of times we're right. Ours: 92% (measured pessimistically).
F1	A single score balancing recall and precision — useful when improving one hurts the other. Ours: 0.923.
False positive	A false alarm — the app marks a pool where there isn't one.
Ensemble	Two or more AI models working as a team, combining their answers.
Fine-tuning	Taking an already-trained AI and giving it extra lessons on your own examples.
Self-distillation	Letting the current AI label data for training the next one — used here only to preserve the solar skill, never for the exam's answer key.
Ground truth	The answer key — points where we know the real answer, used to score the AI.
OpenStreetMap (OSM)	A free world map built by volunteers — the "Wikipedia of maps".
Overpass	The free query service that lets software ask OpenStreetMap questions.
Cadastre / lot-plan	The government's official property-boundary register and a parcel's legal identifier.

Basemap	The background imagery layer of a map.
Zoom 20	A map detail level where one pixel is roughly 15 cm of ground — enough to see a pool clearly.
Lead	A property that looks like a promising customer for a specific service.
CSV	A simple spreadsheet file that opens in Excel.
GeoPackage / QGIS	A standard file for professional mapping software; QGIS is the leading free GIS program.

AUS Backyard Intelligence · build r22 · Two-model YOLO ensemble · recall 92.6% / precision 91.9% / F1 0.923 on a 581-point ground-truth exam · 117 automated tests · All data sources free and open · Full experiment history in docs/accuracy_log.md · Generated from the project's verified state, July 2026.